

Let $\langle M, *, e_M \rangle$ and $\langle T, \Delta, e_T \rangle$ be any two monoids. A mapping $g: M \rightarrow T$ such that for $a, b \in M$

$$g(a * b) = g(a) \Delta g(b) \quad \text{--- (2)}$$

$g(e_M) = e_T$ --- (3) is called monoid homomorphism

Note: If g is onto then (2) $\Rightarrow$ (3)

for any $a \in M$
 $g(a) = g(a * e_M) = g(a) \Delta g(e_M)$
 Since g is onto every $b \in T$ is such that $g(a) = b$ for some $a \in M$.

$$\therefore g(e_M) = e_T$$

$\therefore g$ preserves identity.

More over, if $a \in M$ is invertible and $a^{-1} \in M$ is its inverse then

$$g(a * a^{-1}) = g(e_M) = e_T$$

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$$g(a) \Delta g(a^{-1})$$

$$\text{iii) } g(a^{-1} * a) = g(e_M) = e_T$$

$$\text{ii) } g(a^{-1}) = g(a)^{-1}$$

$$\text{i) } g(a^{-1}) = g(a)^{-1}$$

ii) g preserves inverse

g also preserves commutativity.

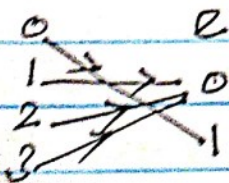
eg: $\langle \mathbb{N}, + \rangle$ - semi group

$S = \{e, 0, 1\}$ with

$\langle S, * \rangle$ - semi group

*	e	0	1
e	e	0	1
0	0	0	0
1	1	0	1

$g: \mathbb{N} \rightarrow S$ gn by $g(0) = 1$ $g(j) = 0$ $j \neq 0$



g is semi group homo

$$g(3+0) = g(3) = 0 = 0 * 1 = g(3) * g(0)$$

$\langle \mathbb{N}, + \rangle$, $\langle S, * \rangle$ monoids, g is not monoid homo
 $\because g(0) \neq e$.

Sub Semigroups and Submonoids.

Defn: Let $\langle S, * \rangle$ — semigrp and $T \subseteq S$.
If the set T is closed under the operation $*$, then $\langle T, * \rangle$ is sub semigrp of S .

|||^{ly} $\langle M, *, e \rangle$ — monoid and $T \subseteq S$
If T is closed under $*$ and $e \in T$, then
 $\langle T, *, e \rangle$ is "sub monoid of $\langle M, *, e \rangle$ "

Eg's: 1. $\langle S, * \rangle$ itself is a trivial sub semigroup of $\langle S, * \rangle$

2. for any $a \in S$, $T = \{a, a^2, a^3, \dots\}$
then $\langle T, * \rangle$ is cyclic semigrp which is sub semigrp of $\langle S, * \rangle$

2. $\langle \mathbb{N}, + \rangle$ semigrp

$T = \{x \mid x \text{ is multiple of a +ve integer } m\}$

$T \subseteq \mathbb{N}$ $\langle T, + \rangle$ is sub semigroup

3. $\langle \{0, 1\}, * \rangle$ is a sub semigrp of $\langle S, * \rangle$ but not submonoid. $S = \{e, 0, 1\}$.

4, $\langle \mathbb{Z}_m, \times_m \rangle$ for $m=5, 6$

The set of invertibles of $\mathbb{Z}_5$ and $\mathbb{Z}_6$

$\{[1], [2], [3], [4]\}$ and $\{[1], [5]\}$ resp

are submonoids.